# James Akl

**Robotics Researcher**

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## COURSEWORK

WPI Transcript | LAU Transcript

Select coursework is tabulated below. Transcripts are linked above to view all courses. Codes are listed to verify courses on transcript.

**Abbreviations** GR = *Graduate* UG = *Undergraduate* WPI = *Worcester Polytechnic Institute* LAU = *Lebanese American University*

| TOPIC GROUP | TOPIC | LEVEL | INSTITUTION | GRADE | COURSE CODES |
|---|---|---|---|---|---|
| **Robotic Systems** | Foundations of Robotics | GR | WPI | A | RBE 500 |
| | Robot Dynamics | GR | WPI | A | RBE 501 |
| | Robot Control | GR | WPI | A | RBE 502 |
| | Biomechanics & Robotics | GR | WPI | A | RBE 520 |
| | Robot & AI Ethics | GR | WPI | – | WR 593 |
| **Pure Mathematics** | Convex Analysis & Optimization | GR | LAU | A | ACM 720 |
| | Real Analysis | UG | LAU | A | MTH 401 |
| | Discrete Structures | UG | LAU | A | MTH 207 |
| **Applied Mathematics** | Computational Inverse Problems | GR | WPI | A | MA 590 |
| | Applied Statistics | GR | WPI | A | MA 511 |
| | Engineering Mathematics | GR | WPI | A | MA 501 |
| | Numerical Linear Algebra | GR | LAU | A | ACM 713 |
| | Numerical Methods | UG | LAU | A | GNE 333 |
| | Probability & Statistics | UG | LAU | A | GNE 331 |
| | Multivariate Calculus (4 courses) | UG | LAU | A | MTH 101, 102, 201, 206 |
| | Differential Equations | UG | LAU | A | MTH 304 |
| | Finite Element Methods | UG | LAU | A | MEE 421 |
| | Optimization | UG | LAU | A | INE 420 |
| **Software Systems** | Advanced Cryptography | GR | WPI | A | ECE 673 |
| | Python Programming | GR | WPI | A | CS 5007 |
| | Object-Oriented Programming | UG | LAU | A | COE 212 |
| | MATLAB Programming | UG | LAU | A | MEE 212 |
| **Management Science** | Entrepreneurship & Innovation | UG | LAU | A | ETR 500 |
| | Engineering Economy | UG | LAU | A | INE 320 |
| | Microeconomics | UG | LAU | A | ECO 201 |
| | Project Management | UG | LAU | B | INE 428 |
| **System Design** | Control Systems | UG | LAU | A | MEE 445, 446 |
| | Instrumentation & Measurements | UG | LAU | B | MEE 391 |
| | Electrical Circuits | UG | LAU | B | ELE 305 |
| | Computer Aided Design | UG | LAU | A | MEE 351 |
| | Engineering Graphics | UG | LAU | A | MEE 211 |
| **Mechanical Systems** | Dynamics & Statics (3 courses) | UG | LAU | A | PHY 111, CIE 200, MEE 241 |
| | Mechanical Vibrations | UG | LAU | A | MEE 442, 443 |
| | Kinematics of Linkages | UG | LAU | B | MEE 341 |
| | Mechanical Design | UG | LAU | B | MEE 422 |
| **Thermal Systems** | Fluid Mechanics | UG | LAU | B | MEE 311, 312 |
| | Thermodynamics (2 courses) | UG | LAU | B | MEE 301, 302 |
| | Heat Transfer | UG | LAU | B | MEE 403, 404 |
| | Internal Combustion Engines | UG | LAU | A | MEE 407 |
| | Refrigeration & Air Conditioning | UG | LAU | B | MEE 515, 516 |
| | Thermal System Design | UG | LAU | B | MEE 414 |
| **Materials Science** | General Chemistry (2 courses) | UG | LAU | A | CHM 101, 200 |
| | Material Properties | UG | LAU | A | MEE 321 |
| | Strength of Materials | UG | LAU | B | MEE 320 |
| | Manufacturing Processes | UG | LAU | B | MEE 332, 333 |